BLUETOOTH INTERFACE & SOFTWARE GUIDE · XR · XR-16

BLE Protocol, Apps, VRChat & Firmware

Everything needed to use PiEEG XR from software: the apps, the Bluetooth service and sample format, conversion to microvolts, a minimal Python reader, PiEEG Server with VRChat OSC, the example dataset and the firmware update.



BLE 5	3 B	250/s	OSC	PiEEG XR
Notify, no pairing	Per channel, 24-bit	Samples per second	VRChat on UDP 9000	Advertised name

01 SOFTWARE OPTIONS

Option	Best for	Runs on	Start
BodyPress	Signals inside VR, VR games	Meta Quest	Meta Quest Store → BodyPress
BodyPress	Quick tests and recording	Android	Google Play → BodyPress
PiEEG Cloud	No headset needed, dashboards, Face Trainer	Chrome / Edge	cloud.pieeg.com → PiEEG XR
PiEEG Server	Recording, LSL, OSC, WebSocket	Windows, macOS, Linux	`--device ironbci8`
VRChat avatar	Avatar reacting to EEG bands	PC + VRChat	`pieeg-server ... --osc`

02 BLE INTERFACE

Advertised name	`PiEEG XR` (the test firmware in the repository advertises `PiEEG_XR`)
Data characteristic	`0000fe42-8e22-4541-9d4c-21edae82ed19` — notify
Pairing	Not needed. Connect directly and subscribe to notifications.
Sample	3 bytes per channel, 24-bit big-endian two's complement; channels in order 1 … N
Channels per sample	N = 8 on PiEEG XR · N = 16 on PiEEG XR-16 (CH1–8 from ADS1299 #1, CH9–16 from ADS1299 #2)
Notification	One or more whole samples; on XR-16 typically 96 bytes = 2 samples × 16 channels
Protocol family	IronBCI — same characteristic as IronBCI-8 and IronBCI-16

03 CONVERTING TO MICROVOLTS

$$\mu V = raw \times 4.5 / (2^{24} - 1) \times 10^{6}$$

raw is the sign-extended 24-bit value. This is the conversion PiEEG Server uses for PiEEG XR (≈ 0.268 µV per count).

NOTE DC offset is normal.

Raw values carry an electrode offset of thousands of µV. Band-pass (e.g. 1–40 Hz) before looking at the signal.

PYTHON

```
import numpy as np
UV = 4.5 / 16777215 * 1e6              # ~0.268 uV per count
N = 8                                  # 16 on PiEEG XR-16

def parse(payload: bytes) -> np.ndarray:
    """notification -> (samples, N) in uV"""
    n = len(payload) // (3 * N) * 3 * N
    b = np.frombuffer(payload[:n], np.uint8).reshape(-1, 3)
    v = (b[:, 0].astype(np.int32) << 16) | (b[:, 1] << 8) | b[:, 2]
    v = np.where(v & 0x800000, v - 0x1000000, v)
    return v.reshape(-1, N) * UV
```


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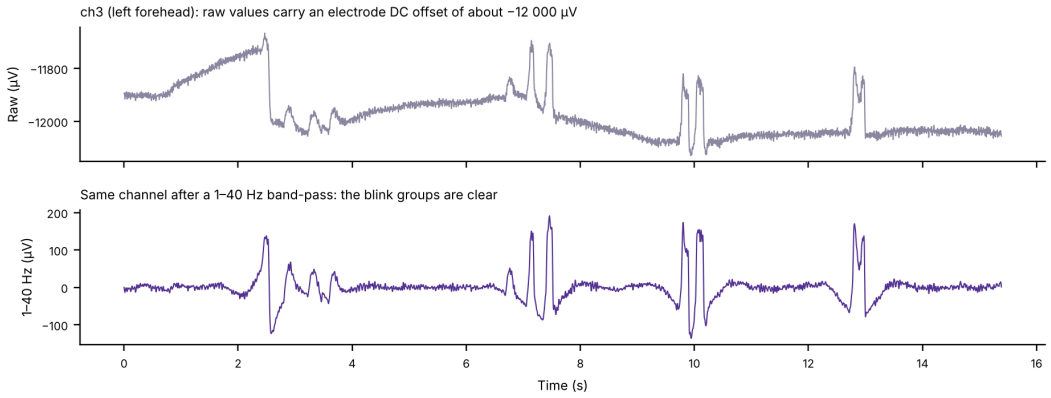


Figure 1 — Datasets/Raw Data EEG Blinking 4-3-2-1.csv, ch3: raw values and the same data after a 1–40 Hz band-pass.

04 MINIMAL READER

PIP INSTALL BLEAK NUMPY

```
import asyncio
from bleak import BleakClient, BleakScanner
CHAR = "0000fe42-8e22-4541-9d4c-21edae82ed19"
is_xr = lambda d, adv: (d.name or "").replace("_", " ").startswith("PiEEG XR")

def on_data(_, data):
    for sample in parse(bytes(data)):        # parse() from section 03
        print([round(v, 1) for v in sample])

async def main():
    dev = await BleakScanner.find_device_by_filter(is_xr, timeout=15)
    async with BleakClient(dev) as client:
        await client.start_notify(CHAR, on_data)
        await asyncio.sleep(10)              # stream for 10 s

asyncio.run(main())
```

05 PIEEG SERVER AND VRCHAT OSC

TERMINAL

```
pip install "pieeg-server[ironbci]"

pieeg-server --device ironbci8                    # scan
pieeg-server --device ironbci8 --ble-name "PiEEG XR"
pieeg-server --device ironbci8 --osc              # VRChat
```

Dashboard at **http://localhost:1617**. The IronBCI driver reads **8 channels**; for PiEEG XR-16 use PiEEG Cloud.

With **--osc** the server sends five floats (0–1, normalised) to VRChat on **UDP 9000**. Enable OSC in VRChat's Action Menu and add the parameters to the avatar in Unity.

OSC address	Band	Use
/avatar/parameters/EEG_Delta	0.5–4 Hz	Drowsiness
/avatar/parameters/EEG_Theta	4–8 Hz	Meditative state
/avatar/parameters/EEG_Alpha	8–13 Hz	Relaxation — easiest to control
/avatar/parameters/EEG_Beta	13–30 Hz	Focus
/avatar/parameters/EEG_Gamma	30–100 Hz	High-frequency activity

OSC is a PC feature: on a Meta Quest, run VRChat on the PC through Air Link or Steam Link. Step-by-step manual: Docs/PiEEG_XR_VRChat_Manual.docx in the repository.

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06 DATASET

File (Datasets/)	Content	Samples
`Raw Data EEG Blinking 4-3-2-1.csv`	Raw, with electrode DC offset	3,805 (≈ 15.4 s)
`Data Band Pass (1-40 Hz) EEG Blinking 4-3-2-1.csv`	Same test, band-pass 1–40 Hz	3,015 (≈ 12 s)

Columns: timestamp (Unix time, s), ch1 … ch8 in µV. Timestamps are coarse (shared by many rows); the raw file averages ≈ 247 samples/s.

07 FIRMWARE UPDATE

Needed if the device does not connect over Bluetooth. You need the **ST-Link V2** from the package and **STM32CubeProgrammer**.

1 Connect

Open the box lid; ST-Link to the SWD header: GND, RST, SWDIO, SWCLK (XR-PIN-004). Mask off the face.

2 Flash the application

STM32CubeProgrammer → ST-LINK → Connect; open the latest `.hex` from Firmware/hex (currently `PiEEG_XR_test.hex`) → Download.

3 Reset

Disconnect the ST-Link and unplug the power for a few seconds. The device advertises as PiEEG XR.

4 BLE stack — only if support asks

Firmware Upgrade Services: `stm32wb5x_BLE_Stack_full_fw.bin` at **0x080CE000** → Firmware Upgrade, then flash the application again.

> **CAUTION A wrong address can disable the radio.**
> Only update the BLE stack when PiEEG support asks you to, with the file and address above.

08 RELATED DOCUMENTS

XR-DS-001 Product Datasheet	**XR-MD-002** Mechanical & Assembly Drawing	**XR-PCB-003** PCB & Manufacturing Specification
XR-PIN-004 Mask, Electrode & Connector Guide	**XR-QSG-005** Quick Start & Safety Guide	

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